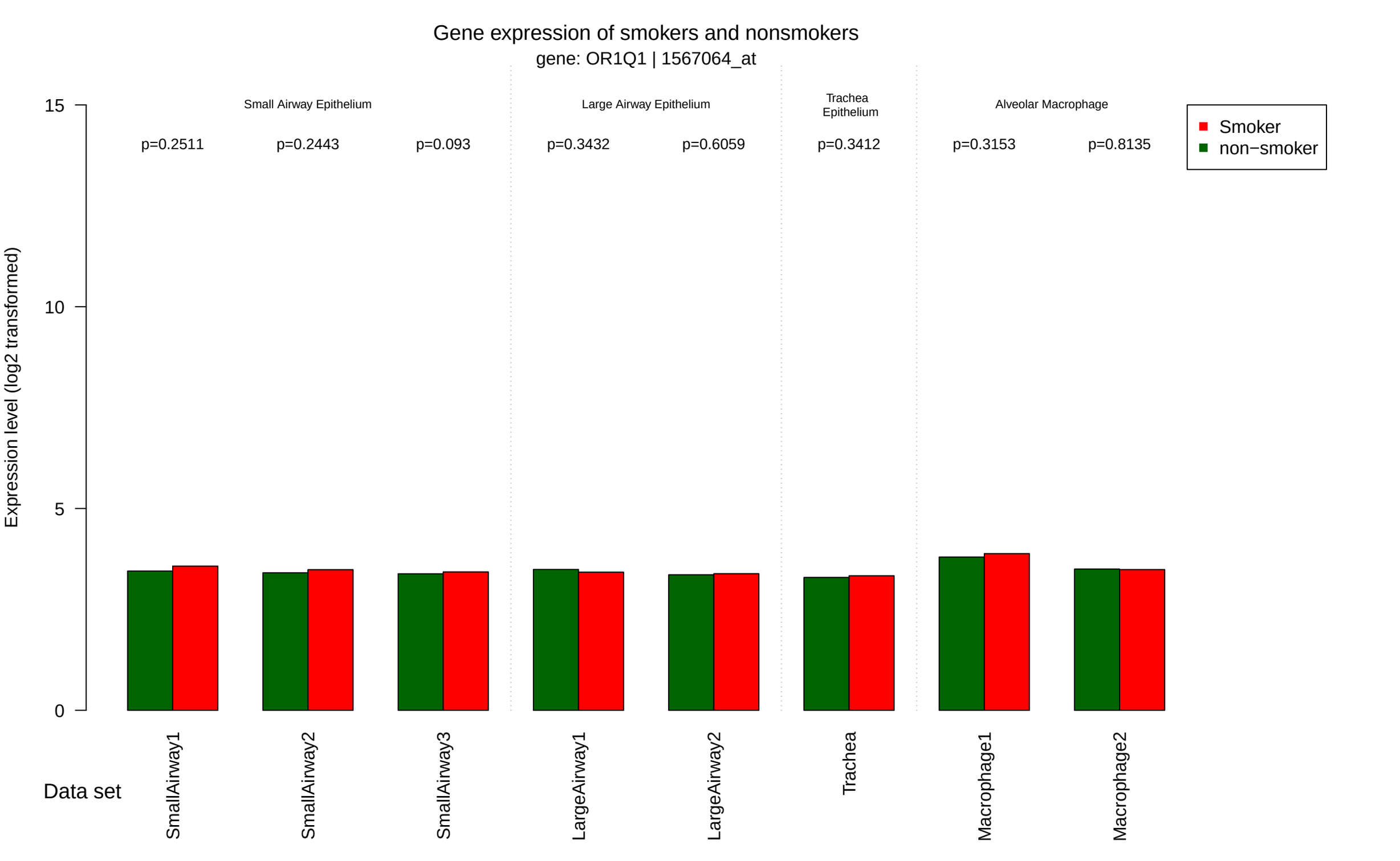
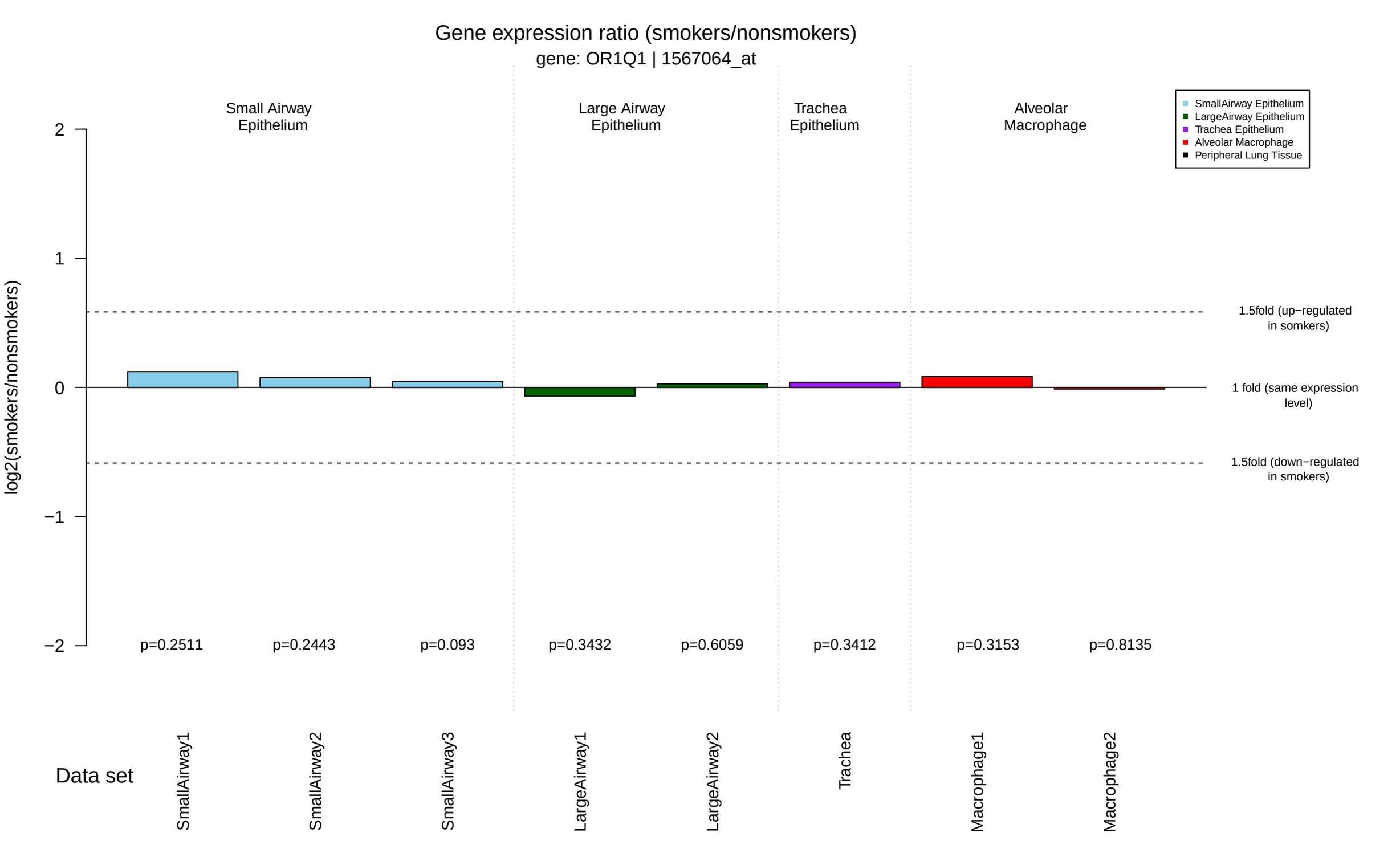
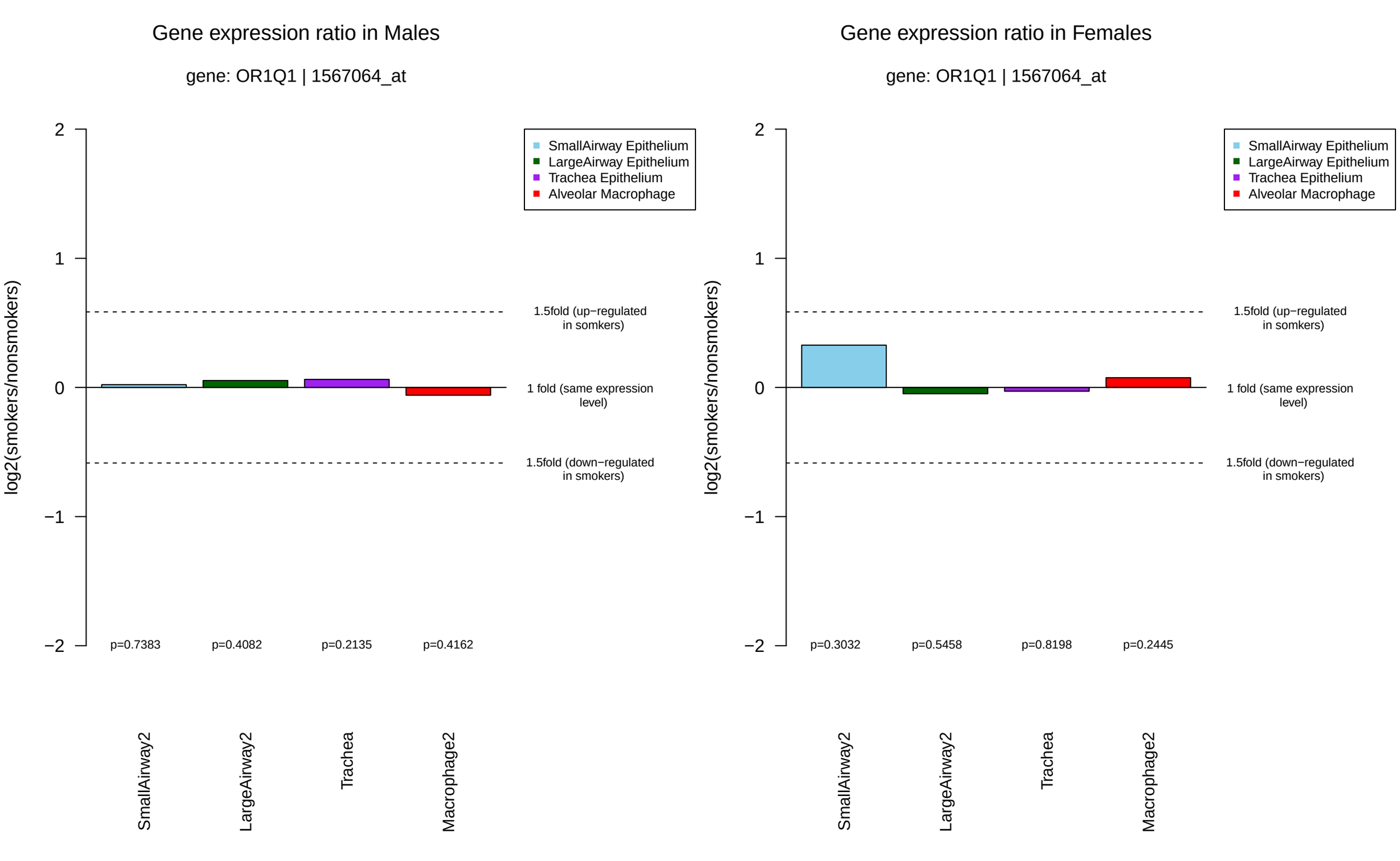
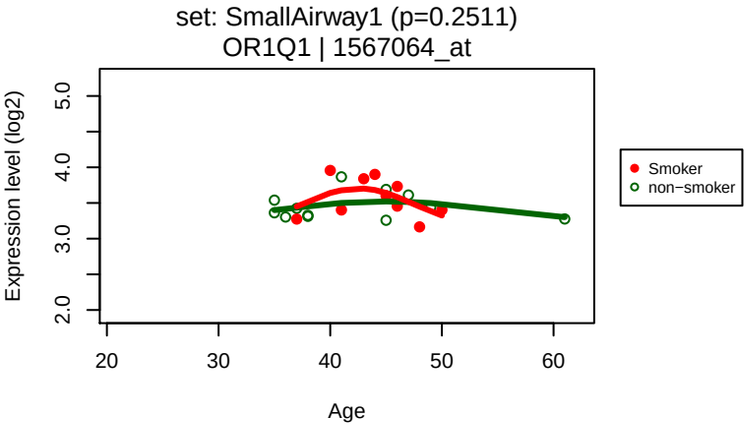
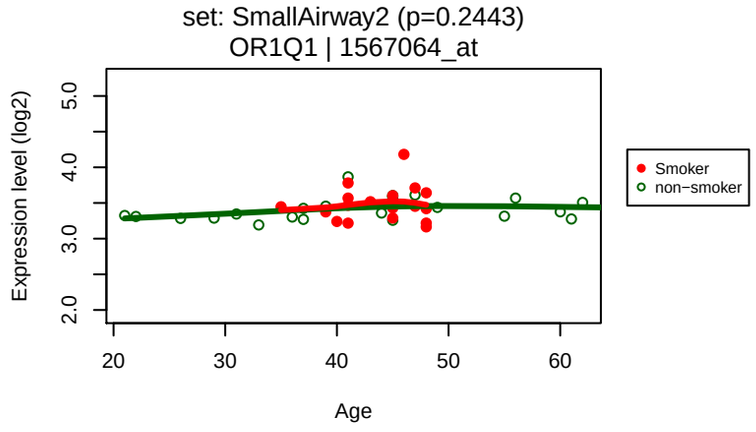
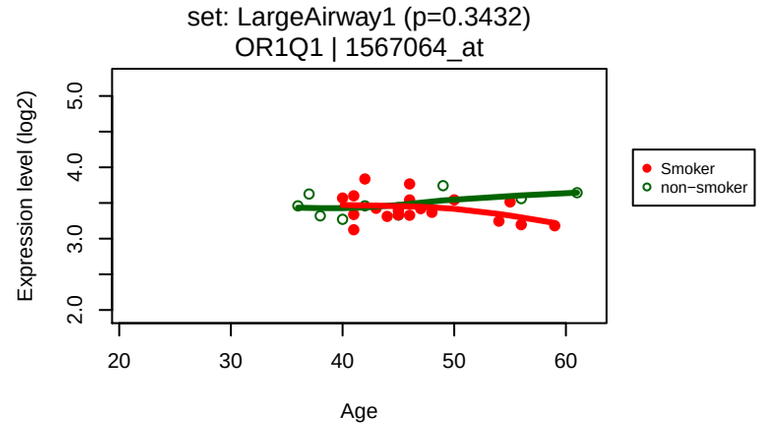

Gene expression of smokers and nonsmokers
gene: OR1Q1 | 1567064_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2511
p=0.2443
p=0.093
p=0.3432
p=0.6059
p=0.3412
p=0.3153
p=0.8135
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

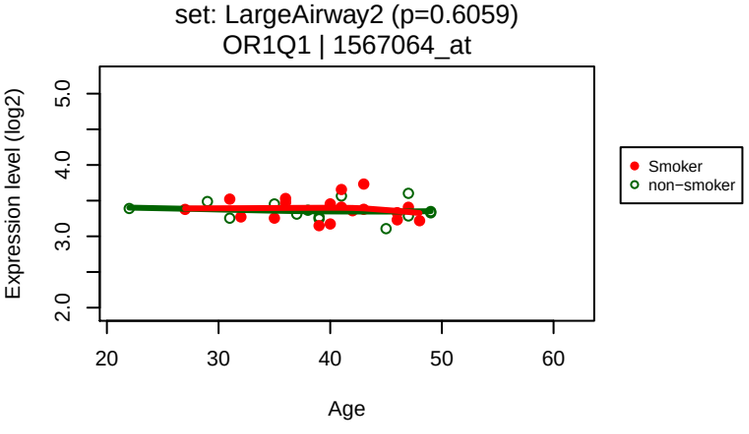

Gene expression ratio (smokers/nonsmokers)
gene: OR1Q1 | 1567064_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2511
p=0.2443
p=0.093
p=0.3432
p=0.6059
p=0.3412
p=0.3153
p=0.8135
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

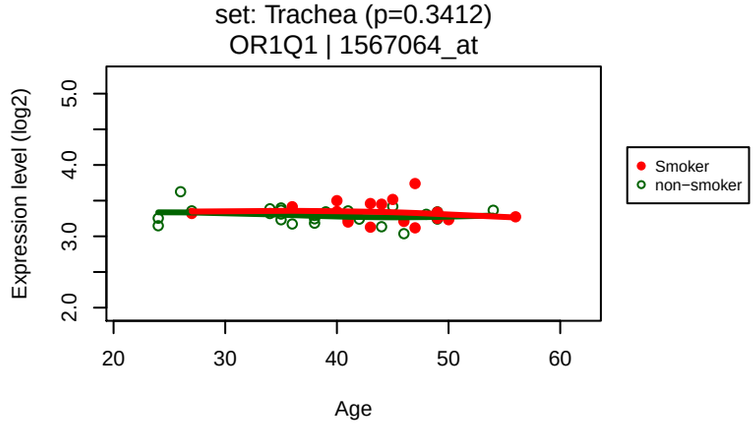

Gene expression ratio in Males
gene: OR1Q1 | 1567064_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7383
p=0.4082
p=0.2135
p=0.4162
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR1Q1 | 1567064_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3032
p=0.5458
p=0.8198
p=0.2445
SmallAirway2
LargeAirway2
Trachea
Macrophage2

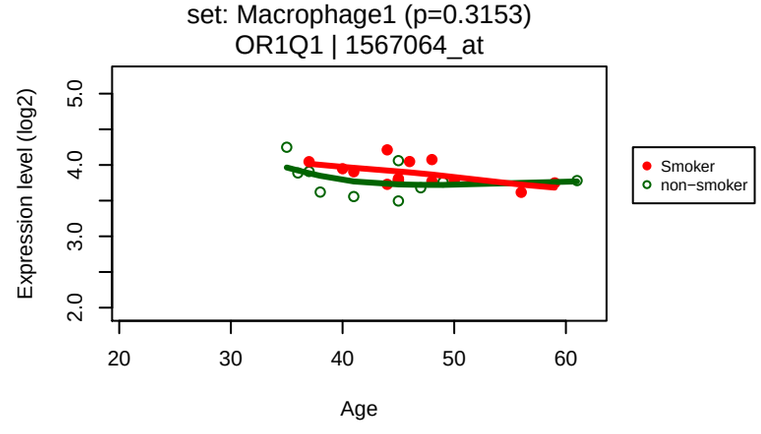

set: SmallAirway1 (p=0.2511)
OR1Q1 | 1567064_at
Expression level (log2)
Age
Smoker
non−smoker

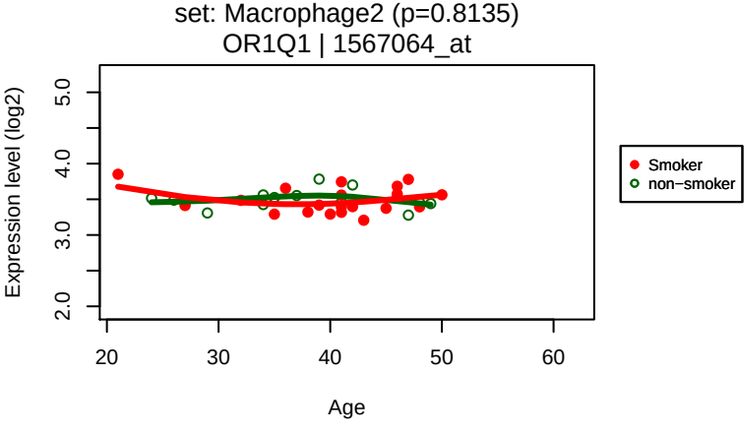

set: SmallAirway2 (p=0.2443)
OR1Q1 | 1567064_at
Expression level (log2)
Age
Smoker
non−smoker

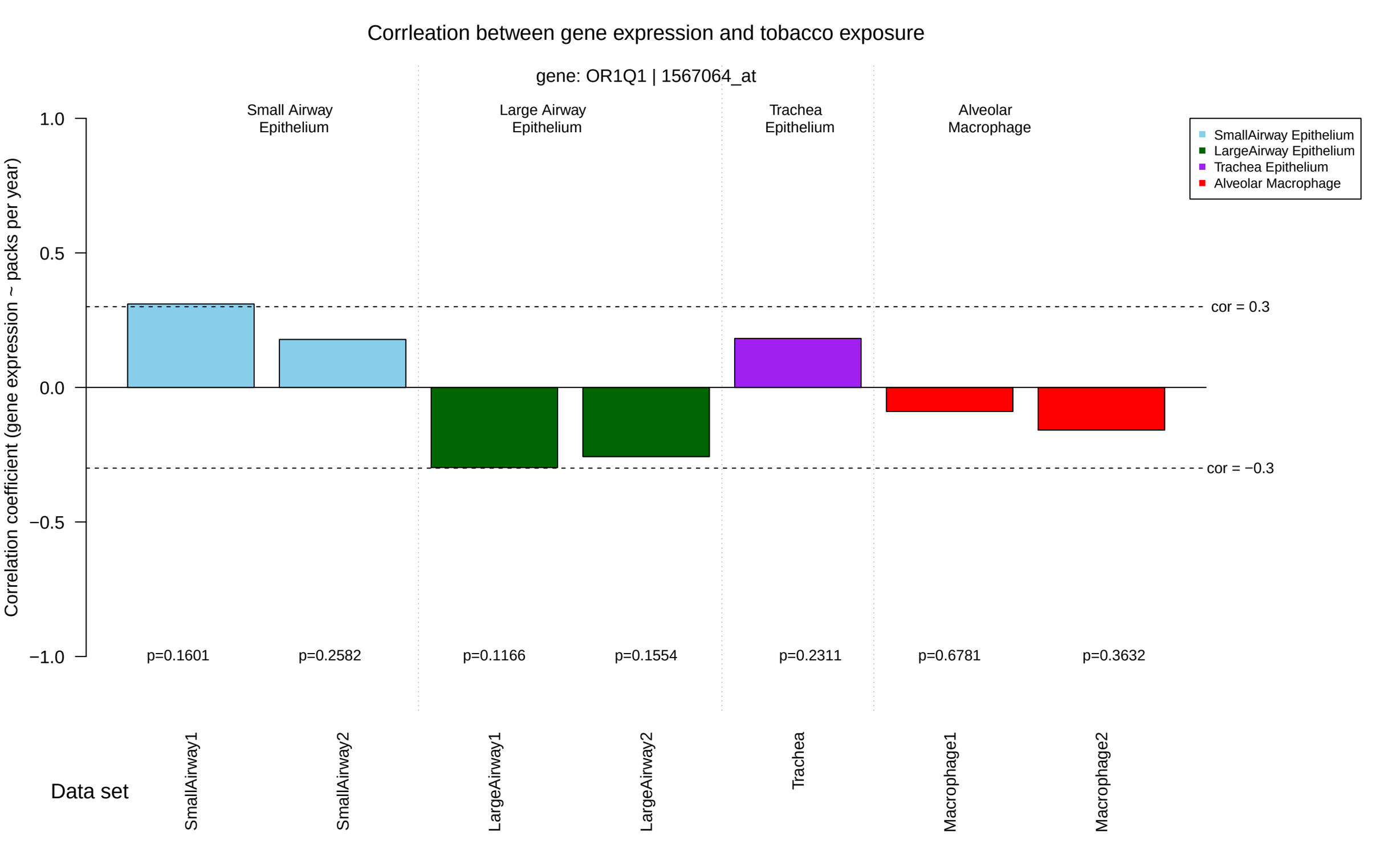

set: LargeAirway1 (p=0.3432)
OR1Q1 | 1567064_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6059)
OR1Q1 | 1567064_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3412)
OR1Q1 | 1567064_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3153)
OR1Q1 | 1567064_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8135)
OR1Q1 | 1567064_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR1Q1 | 1567064_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1601
p=0.2582
p=0.1166
p=0.1554
p=0.2311
p=0.6781
p=0.3632
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2